

AVAW_CV-Waste: An Offline Computer Vision Annotation Tool for Waste Management Research

Gerald Koinig^{1¶} and Bojan Lorber¹

¹ Montanuniversität Leoben, Austria ¶ Corresponding author

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1 Summary

AVAW_CV-Waste is a free and open-source offline desktop application for creating and correcting image annotations used to train object detection models. The software allows users to draw, edit, move, resize, copy, and manage bounding boxes around objects in images, with outputs compatible with YOLO-based artificial intelligence workflows, specifically the YOLO PyTorch TXT annotation format commonly used by YOLO11, YOLOv8, YOLOv5, and related computer vision models.

The software also supports the integration of existing YOLO models for semi-automatic annotation generation, reducing the amount of manual labelling required during dataset preparation. The application was designed specifically for waste management, recycling, and material sorting research, where datasets often contain highly domain-specific object classes and cannot easily be processed using generic or cloud-based annotation platforms.

Because the software operates entirely offline, it can be deployed directly at industrial facilities or research sites where internet access may be limited or unavailable. Local processing additionally guarantees data sovereignty, an important requirement for industrial research collaborations involving sensitive operational data.

The primary goal of AVAW_CV-Waste is to simplify and accelerate dataset creation for non-specialist users, including students, laboratory staff, recycling operators, and industrial research partners. By reducing the technical complexity of annotation workflows, the software supports the development of custom machine learning models for applications such as waste classification, contaminant detection, scrap sorting, and sensor-based recycling research.

2 Statement of need

The AVAW_CV-Waste Tool was developed as a free and open-source alternative that can be adapted to specific scientific workflows, especially in waste management and recycling research, where open-source data is scarce to non-existent.

In this domain, images of particles with contamination, deformation and on conveyor belts, specific to the actual re-search question regarding the sorting and classification of waste particles are necessary.

Thus, waste datasets often require highly specific object classes, such as plastics, paper, scrap-types, hazardous waste, or mixed waste fractions in environments relevant to the waste management domain.

Existing tools may not easily support these specialized categories or the frequent changes that occur during research projects and often prohibit or disincentivise the use of custom pre-trained models, which may speed up the annotation process.

39 Lastly, since many existing tools are cloud based, data sovereignty may not always be guaranteed
40 - an increasingly precarious issue when working with industrial partners in waste management
41 research projects heeding to data often being recorded directly in respective material recovery
42 facilities.

43 This tool allows users to define their own custom annotation classes, load and use their own
44 models for automatic predictions, work entirely offline without requiring cloud services organize
45 datasets according to their own folder structures and workflows and quickly review, edit, and
46 correct annotations at no cost.

47 The software primarily targets researchers and industrial practitioners working in waste
48 management, recycling, circular economy technologies, and sensor-based sorting applications.

49 **3 State of the field**

50 The field of annotation software allows for several pre-build alternatives.

51 However, as we noticed in many of our research projects, these tools' reliance on online
52 annotation with the requirement to upload the dataset and related annotations into a cloud
53 service often clashes with industrial partners need for data sovereignty.

54 In theory one can build a self-hosted version and move docker images to offline machines.

55 But one would need advanced technical skills and additional resource and time for that.

56 Further, this approach would eliminate the application of pre-trained models to semi-
57 automatically segment and annotate objects in one's dataset.

58 However, this automated annotation is oftentimes limited paid plans to alleviate usage limits
59 and the available models provided by cloud service in question, prohibiting using one's own
60 pre-trained waste management domain specific models for aiding the annotation process.

61 In conclusion, the emerging research field of using machine learning models for waste
62 management classification tasks needed an open source, free of cost, offline and adaptable
63 tool that allows for modular expansions and use of domain specific detection models that also
64 ensures data-sovereignty and is not reliant on online services once installed.

65 The tool therefore fills a niche between generic cloud annotation platforms and highly customized
66 in-house software solutions used in industrial recycling research.

67 **Software design**

68 While we are aware that basic code for software like this can rapidly be developed using
69 generative ai, extensive testing under real research scenarios with researchers of varying degrees
70 of technical prowess can't be offloaded to GenAI.

71 Thus, the UI and UX of this tool were subject to constant iteration and improvement, spanning
72 multiple years of research projects in this emerging field in waste management research. It
73 could therefore incorporate criticism and feedback from students during lectures, researchers
74 and workers that annotated thousands of images using this tool. Ongoing usage and the
75 everchanging nature of research projects in general and Waste Management related sorting tasks
76 in particular will inevitably uncover further optimisations and adaptations. The use of Python
77 itself and the common UI framework tkinter as well and an object oriented approach to the
78 tools makeup is aimed at enabling customisation and extension by other researchers working
79 in waste management and recycling domains. Since Python is widely adopted in machine
80 learning and computer vision research, the software integrates naturally into existing scientific
81 workflows.

82 Several design trade-offs were made to ensure broad hardware compatibility and stable operation
83 on lower-performance research machines, while still allowing for quality of life features expected
84 from a tool like this. Lightweight rendering and simplified interaction workflows were prioritized
85 over computationally expensive visualization features.

86 Overall, we chose Python and tkinter as the underlying architecture to leverage both the wide
87 support for Python applications, its ease of use for other waste management researchers to
88 adapt the tool for their purposes and of course the widespread adoption of Python as the
89 programming language for rapid development of machine learning tools.

90 Research impact statement

91 AVAW_CV-Waste has already been successfully used in the development of scientific datasets
92 and in the publication of research related to waste detection, classification, and recycling
93 workflows.

94 AVAW_CV-Waste addresses a major bottleneck in the development of computer vision systems
95 for waste management and recycling research, namely the creation of annotated training
96 datasets.

97 The software is currently being used within multiple recycling and circular economy lighthouse
98 research projects, including:

- 99 ▪ KiRAMET: AI-based recycling of metal composite wastes,
- 100 ▪ StraTex: Sorting and processing strategies for used textiles,
- 101 ▪ greenPLAST-food: Green plastic recycling for food contact materials, and
- 102 ▪ Scarpa: Strategic recycling of footwear.

103 Within these projects, the software has already been applied in research workflows involving
104 post-consumer textiles, lightweight packaging waste, and post-shredder scrap, where large
105 quantities of annotated image data are required for training and evaluating object detection
106 models.

107 The created datasets formed the basis for scientific publications showcasing the applicability of
108 machine learning methods for sorting heterogeneous waste streams and reducing manual sorting
109 requirements ("AI Pipeline for Sensor Based Sorting of Shredder Scrap Towards Green Steel
110 Production," 2026; *Deep Learning Approaches for Classification of Copper-Containing Metal
111 Scrap in Recycling Processes*, 2026; *Detection of Copper-Containing Scrap in a Post-Shredder
112 Fraction with Machine Vision and Artificial Intelligence Towards Green-Steel Production*, 2026;
113 *GreenPLAST-Food*, 2026; *KiRAMET*, 2026; *Kiramet*, 2026; "Robust YOLO-Based Ejection
114 of Copper-Containing Particles in Heavily Corroded Scrap Towards Green-Steel Production,"
115 2026).

116 AI usage disclosure

117 Generative artificial intelligence tools were not used for the architectural design or core software
118 implementation of AVAW_CV-Waste.

119 ChatGPT (OpenAI) was used during software development for documentation support,
120 generation of docstrings, bug-fixing assistance, and minor inline code modifications. For this
121 manuscript, generative AI tools were used only for grammatical improvements and formatting
122 support, including conversion of manuscript content from .docx to Markdown syntax.

123 All generated content and code modifications were manually reviewed and validated by the
124 authors.

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